

CS3101: Programming and Data Structures I Class Test Extra (Slot 1)
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Starts: 29/11/2025 at 10 AM IST (Duration: 1 hour 45 mins)

Total marks: 25

Instructions

- All programs should be compatible with GNU C.
- Late submissions will attract marks deduction.
- You are advised to compile the code before submitting it to WeLearn. If a code does not compile at our end, marks will be deducted for the problem.
- Each program should follow a strict naming convention: **QNo.c** (e.g. Q1.c, Q2a.c etc.). Programs not adhering to the convention will not be evaluated.
- All codes (.c files) should be submitted in a single zipped folder to WeLearn.
- You should put appropriate comments in the code.

1. **(Marks: 10)** Write a program to implement a function `IsSubstr(s1, s2)` which checks if string *s1* is a substring of another string *s2* or not. E.g., *ayan* is a substring of *Dwaipayan*, but *maya* is not. You may call `IsSubstr()` from `main()` passing as arguments, string pairs (*ayan*, *Dwaipayan*) (should return 1) and (*maya*, *Dwaipayan*) (should return 0), to test the correctness of the function. You are not allowed to use any built-in function for the same that can replicate `IsSubstr(s1, s2)`.
2. **(Marks: 15)** In a dance reality show *Dancistan*, there are five contestants: *Dev*, *Ankush*, *Allu*, *Nora* and *Tamannah*. There are three rounds where three judges (*Madhuri*, *Prabhu*, and *Mithun*) score each of the participants, and the sum of scores of the three judges is considered for each contestant.
 1. **Round 1:** The lowest scorer (based on the sum of three judges for this round) gets eliminated; the top 4 contestants remaining
 2. **Round 2:** The lowest scorer among the top 4 (round 1 score carried forward + round 2 score, which is again the sum of three judges for round 2) gets eliminated; the top 3 contestants remaining
 3. **Round 3:** The lowest scorer among the top 3 (round 1 score + round 2 score carried forward + sum of three judges for round 3) gets eliminated, to set up the finale between the top 2.

However, one of the three eliminated contestants returns for the finale through public voting as a **wildcard entry**.

At the finale, the top 2 and the wildcard entry compete, and the winner, the first runner-up, and the second runner-up are decided *only on the finale performance*, again on the sum of three judges.

Simulate the whole process through C implementation: take scores (and wildcard entry) as input, print the eliminated participants at the first three rounds, and finally print the three position holders for the finale. **Assume that the judges input their scores (at all rounds and in the finale) through the terminal, and the public voting choice is also through the terminal.** You may use only local variables; file handling is not necessary.